



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS

PROJECT PROPOSAL

ON

BAGH CHAL GAME



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Table of contents:

TITLE	Page No.
1. Introduction	1
1.1 Introduction to Bagh Chal	1
1.2 Key technologies	2
1.3 Key Terminologies for artificial intelligence integration	3
2. Objectives	5
3. Existing system	6
4. Proposed system	7
5. Methodology	8
6. Project Scope	9
7. Project Schedule	10

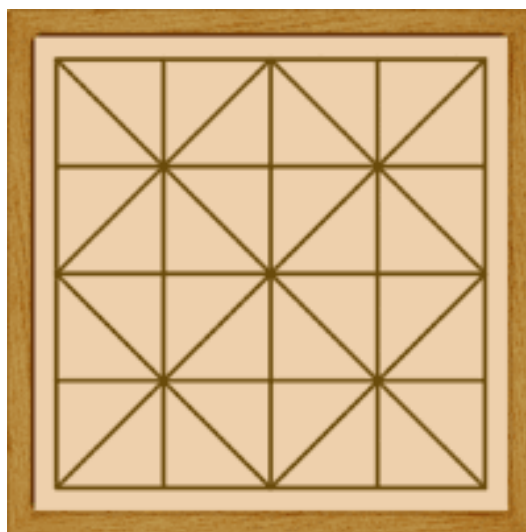
1. Introduction

In this modern era, youth and youngsters are quickly transitioning into modern games, while traditional games have started declining in popularity. One of these old games is Bagh-Chal. Migrating Bagh Chal to a digital platform not only preserves this rich cultural asset but also makes it globally accessible. While web-based or mobile adaptations exist, they are often built clumsily, not meeting standards of modern strategy games like chess. This project attempts to improve the perspective of the younger generation towards this old game, by modernizing it as necessary.

This project aims to bridge the gap between cultural preservation and high-performance software engineering by developing a native desktop implementation of Bagh Chal. The application is built entirely using C++ to ensure optimal memory management and execution speed. The graphical user interface (GUI), hardware-accelerated 2D rendering, and real-time input handling (such as mouse clicks and drag-and-drop mechanics) will be powered by SDL3 (Simple DirectMedia Layer 3).

1.1 Introduction to Bagh Chal

Bagh Chal is a traditional, strategic, two-player board game deeply rooted in the cultural heritage of Nepal. Unlike symmetric games like chess or checkers, Bagh Chal features an asymmetric gameplay dynamic. One player controls four powerful Tigers (Bagh) with the objective of capturing the opponent's pieces, while the other player commands twenty Goats (Bakhra) aiming to encircle and immobilize the tigers. The game unfolds on a specific 5x5 grid interconnected by orthogonal and diagonal lines, demanding high spatial awareness and tactical foresight from both sides.



Bagh Chal game board

1.2 Key Technologies

The software tools that we used to prepare this project are listed below:

Programming language : C++

C++ is a high performance, highly versatile programming language that works well with both procedural and Object Oriented Programming paradigms. We chose this language because:

1. It is very relevant as a part of our curriculum.
2. Object Oriented Architecture - It allows complex game elements (like the board state, piece mechanics, and players) to be cleanly structured into reusable classes and modules.
3. It is the industry standard language for game development, making it relevant for future opportunities.

Compiler : GCC

It is an open-source compiler system produced by the GNU Project. This was chosen because:

1. Very easy to access, widely used, and reliable.
2. Cross platform support - Any system can install and run this compiler using ports such as MinGW (for Windows).
3. Easy to integrate - It integrates flawlessly with VS Code's C/C++ extensions, the built-in terminal, and standard build tools (like Makefiles or CMake), creating a smooth process from writing code to launching your SDL3 window.

Library : SDL3 (Simple DirectMedia Layer 3)

SDL3 is a cross-platform development library designed to provide low-level access to audio, keyboard, mouse, joystick, and graphics hardware. We chose this for the following reasons:

1. It is the industry standard library for 2D game development.
2. Hardware acceleration - It features a 2D rendering API that utilizes hardware acceleration, ensuring clean and smooth flow of the program.

3. Cross platform and modern - As the latest iteration of SDL, it offers modernized API handles, better window management, and native cross-platform support out of the box.

Library: SDL (Standard Template Library)

A powerful collection of built-in C++ template classes providing standard data structures (such as vectors, stacks, and maps) and pre-optimized algorithms. We chose this because:

1. Dynamic Game Tree Management - The Minimax algorithm requires generating and traversing large game trees of future board states.
2. Performance-Critical Math Utilities - Native functions within the <algorithm> library (such as std::max and std::min) are leveraged directly inside the Alpha-Beta pruning logic to quickly evaluate score weights, keeping the AI's execution pipeline fast and lightweight.

IDE : Visual Studio Code (vscode)

VSCode is a lightweight, highly customizable, open-source source-code editor developed by Microsoft that supports a large number of extensions. We chose this because:

1. It consumes significantly fewer system resources than heavy IDEs.
2. Customizability and git integration - Its built-in terminal and native Git integration simplify compiling, debugging, and team collaboration.
3. Extension ecosystem - It has easy access to a large number of community-built extensions that improve the coding environment.

1.3 Key Terminologies for Artificial Intelligence Integration

- **Minimax Algorithm:** A recursive, decision-making algorithm widely utilized in artificial intelligence for turn-based, two-player games. It evaluates future game states by assuming both players will play optimally. The algorithm calculates the best possible move by minimizing the potential loss in a worst-case scenario (minimizing the maximum loss).
- **Alpha-Beta Pruning:** An advanced optimization technique applied to the Minimax algorithm. It drastically reduces computational overhead by "pruning" (bypassing) branches of the game tree that are mathematically guaranteed to yield worse outcomes than previously evaluated paths. This ensures faster AI response times without degrading the quality of the final decision.

- **Game Tree:** A directed graph data structure used to represent all possible future board states. In this architecture, each "node" represents a specific layout of the 5x5 Bagh Chal board, and each connecting "edge" represents a legal move that transitions the board from one state to the next.
- **Heuristic Evaluation Function:** A specialized algorithm used to score the strategic value of a given board state when a definitive win or loss has not yet occurred. For Bagh Chal, this function quantifies variables such as the remaining number of goats, the mobility (available jump paths) of the tigers, and the defensive clustering of the goats.
- **Depth Limit (Ply):** The maximum number of future turns (plies) the bot is permitted to simulate. Because the number of possible board combinations grows exponentially with each turn, a strict depth limit prevents the software from exhausting system memory and CPU resources, ensuring the game runs smoothly on standard hardware.
- **Zero-Sum Game:** A competitive mathematical model where one player's strategic advantage results in a mathematically equivalent disadvantage for the opponent. Bagh Chal natively fits this model; an action such as a goat successfully trapping a tiger inherently subtracts from the tiger player's win condition while advancing the goat player's win condition.

2. Objectives

We chose to go ahead with this project with the following objectives in mind:

To design an interactive graphical user interface (GUI):

To utilize SDL3 for rendering a visually clear 5x5 board, drawing tokens, and handling real-time user inputs like mouse clicks and piece dragging.

To develop a local multiplayer architecture:

To create a seamless, turn-based Player-vs-Player (PvP) system that cleanly manages turn switching and updates the board state accordingly.

To implement an Artificial Intelligence opponent:

To integrate a single-player PvE (Player vs. Environment) mode by developing a decision-making bot engine. This engine will utilize the Minimax algorithm, optimized with Alpha-Beta pruning, to efficiently evaluate game trees and calculate optimal strategic moves in real-time without compromising the application's memory or runtime performance.

To apply Object-Oriented Programming (OOP) principles:

To build a modular, well-structured codebase that cleanly separates the frontend rendering engine from the backend game logic, ensuring efficient memory management and high performance.

To implement the core game logic:

To code a rule engine in C++ that strictly enforces Bagh Chal's unique mechanics, including valid tiger jumps, goat placements, and win/loss condition detection (trapping all tigers or capturing 5 goats).

3. Existing System

At present, Baaghchal is available in different digital versions on mobile platforms, where users can play as tigers or goats in one-player or two-player offline modes. Some of these apps also include online play, AI opponents, and side selection features, which show that the game has already been adapted for digital devices.

There are also existing AI-based Baaghchal projects that use minimax and alpha-beta pruning for bot moves. These systems demonstrate that the game logic can be automated and that the bot can make strategic decisions based on the current board state.

These existing apps and projects provide useful examples of digital Baaghchal gameplay, mode selection, and AI-based move generation. My proposed project follows the same game concept but implements it using SDL3 with a custom screen system and a bot built using minimax and alpha-beta pruning.

4. Proposed System

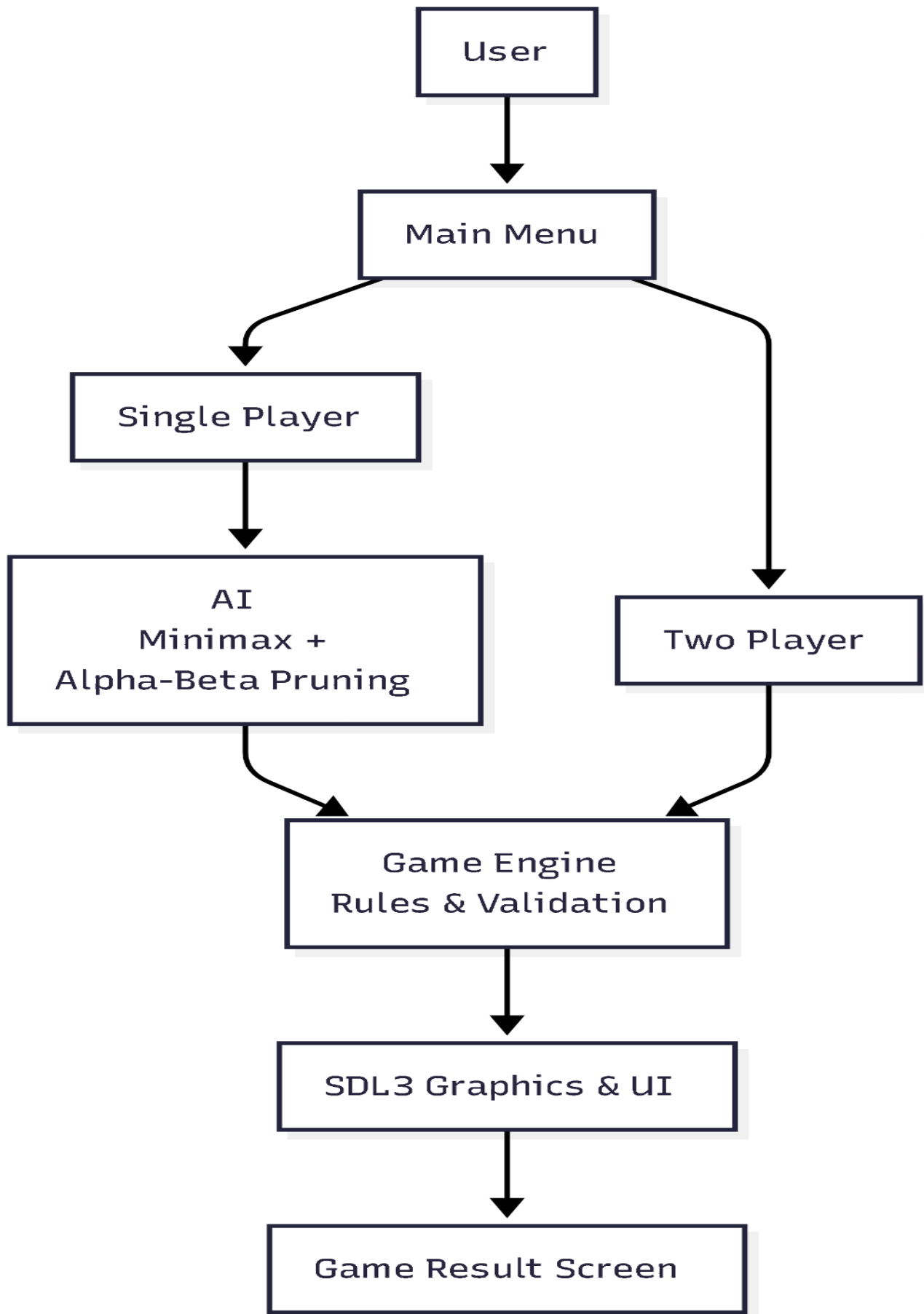
4.1 Description

BaghChal is a traditional two-player strategy board game that originated in Nepal. It is played on a 5×5 board between four tigers and twenty goats. The objective of the tigers is to capture at least five goats, while the objective of the goats is to trap all four tigers so that they cannot move. The game requires strategic thinking and careful planning from both players.

The proposed project is a desktop-based implementation of the BaghChal game developed using C++ and SDL3. It provides both Single Player and Two Player modes. In Single Player mode, the computer opponent uses the Minimax algorithm with Alpha-Beta Pruning to make intelligent decisions and provide a challenging gameplay experience.

The game includes multiple screens such as the Main Menu, Mode Selection, Game Screen, Help Screen, Pause Screen, and Result Screen. It implements all official game rules, including goat placement, tiger movement, goat movement, capture rules, move validation, turn management, and winner detection. The project aims to preserve the traditional Nepali game while providing an attractive, interactive, and user-friendly digital gaming experience.

4.2 System Block Diagram



5. Methodology

The development of the BaghChal Game will follow a systematic software development approach consisting of planning, design, implementation, testing, and deployment. Each phase ensures that the game is developed correctly and meets the project objectives.

1. Requirement Analysis

The first phase involves studying the rules of the traditional BaghChal game and identifying the functional and non-functional requirements. The required features such as Single Player mode, Two Player mode, AI opponent, game screens, move validation, and winner detection are identified and documented.

2. System Design

In this phase, the overall architecture of the game is designed. Different game screens such as the Main Menu, Game Screen, Help Screen, and Result Screen are planned. The game board structure, player interactions, and AI integration are also designed before implementation.

3. Implementation

The game is developed using C++ and SDL3. The game board, user interface, game logic, and screen navigation are implemented. The Single Player mode uses the Minimax algorithm with Alpha-Beta Pruning to calculate the best possible moves for the computer opponent, while the Two Player mode allows two users to play on the same computer.

4. Testing

After implementation, the game is thoroughly tested to ensure that all features work correctly. The testing process includes verifying game rules, validating player moves, checking AI decision-making, identifying bugs, and ensuring smooth screen transitions and gameplay.

5. Deployment and Documentation

After successful testing, the final version of the game is prepared for use. User documentation and the project report are completed, and the application is packaged for submission.

6. Project Scope

The scope of this project is to develop a complete desktop version of the traditional Nepali board game **BaghChal** using **C++** and **SDL3**. The project focuses on creating an interactive, user-friendly, and intelligent game that follows all the official rules of BaghChal. The game will allow users to play either against another player or against the computer.

The main features included in the project are:

- Development of a graphical user interface (GUI) using SDL3 for displaying the game board, pieces, menus, and other screens.
- Implementation of Single Player mode, where the computer acts as the opponent using the Minimax algorithm with Alpha-Beta Pruning to make intelligent decisions.
- Implementation of Two Player mode, allowing two users to play on the same computer.
- Creation of multiple game screens, including the Main Menu, Mode Selection, Game Screen, Help/Instructions Screen, Pause Screen, and Game Result Screen.
- Implementation of all BaghChal game rules, including goat placement, tiger movement, goat movement, tiger captures, legal move validation, and winner detection.
- Automatic checking of valid and invalid moves to ensure fair gameplay.
- Display of game status, player turns, captured goats, and the final winner.
- Smooth navigation between different screens and responsive user controls.
- Well-structured and object-oriented program design for easier maintenance and future improvements.

7. Project Schedule

Week	Task Description
i	Study the rules of BaghChal, gather project requirements, prepare the project proposal, design the system architecture, and create the user interface layout for different game screens.
ii	Develop the game framework using SDL3, implement the game board, create the Main Menu, Game Screen, Help Screen, and Result Screen, and handle user input and navigation.
iii	Develop the game framework using SDL3, implement the game board, create the Main Menu, Game Screen, Help Screen, and Result Screen, and handle user input and navigation.
iv	Implement the complete game logic, including goat placement, tiger movement, capture rules, move validation, turn management, and winner detection.
v	Integrate all modules, perform testing and debugging, improve the user interface and performance, prepare the project documentation, and complete the final report and presentation.